

Quantum API Mermin Pérez

Documentation Bundle

Repository documentation

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1 Overview

This document captures the Markdown and LaTeX-oriented documentation bundle for the Julia workspace. The source content is aligned with the local Documenter setup and focuses on matrix-valued distributions and the solver workflow.

2 Matrix-variate Wishart workflow

The matrix-variate Wishart workflow follows the model

$$X \sim \mathcal{W}_p(n, \Sigma)$$

where $p = 8$ and Σ is the positive-definite scale matrix.

- Build a scale matrix Σ .
- Construct the distribution as `Wishart(9,  $\Sigma$ )`.
- Draw samples with `rand`.
- Normalize the sample by its trace.
- Evaluate the score with `logpdf`.

3 Distributions references

The project documentation is anchored to the JuliaStats references:

- <https://juliastats.org/Distributions.jl/stable/>
- <https://juliastats.org/Distributions.jl/stable/matrix/>
- <https://juliastats.org/Distributions.jl/stable/mixture/>
- <https://juliastats.org/Distributions.jl/stable/fit/>

4 Solver reference

The repository also includes a 25×25 Sudoku solver pattern that can be documented and rendered as part of the narrative site. The constraint system is based on the usual Latin-square rules:

$$\forall r, c: \quad \text{cell}(r, c) \in \{1, \dots, 25\}$$

5 Minimal Julia example

```
using Distributions
using LinearAlgebra

M = Matrix{Float64}(I, 2, 2)
println(tr(M))
```